

G. Tardini and E. Fable

ASTRA–8.0

Addendum with changes and new features

April 2023

MAX-PLANCK-INSTITUT FÜR PLASMAPHYSIK

GARCHING BEI MÜNCHEN

Introduction

Astra 8 is essentially a "new" code with respect to old Astra versions. The Code is now recast in F90, using modules and avoiding at all commons. This makes the code much more efficient to be developed and coupled to other frameworks.

The compiler is now much cleaner, and many systems are already included in the platform list. A user has no difficulty including new ones in case.

Below this is written in details.

Contents

1	New compiler and code language	4
2	Changes in the model file syntax	5
3	Equilibrium solvers	6
4	Generate the machine configuration grid for FEQIS	6
5	Additional routines for the flight simulator	7
6	The new equilibrium solver FEQIS	7
6.1	Input files	8
6.2	Derivation of the equation implemented in Astra-7.0	8
6.3	Evaluation of the radial current and consequences on momentum transport	9
6.4	Internal variables	14
6.5	Dedicated control parameters	15

1 New compiler and code language

The code structure is now the following.

Once cloned, the code comes with a 2-layers structure. Let us suppose the code has been cloned in a directory called "a8code". Then the 2 layers are:

- 1) a8code/
- 2) a8code/repo_user_area/

"a8code" contains the source code directory src/, in which for, feqis, spider, nbi are contained.

"a8code/repo_user_area" contains the working directories with the pre-installed formulas, subroutines, and example model and experimental files. In addition, it contains the directories: "main" - main ASTRA program astra_main.f90 and subroutine stepup.f90, and text files containing the list of all the native variables of various types.

The command "install.sh" will copy from "repo_user_area" to the main directory, all the sub-directories contained there. As such, the user should eventually back-up any changes done to the pre-installed files. The rationale behind this is that the copied directories like sbr/, fnc/, fml/, xpr/, ... can now include user-defined files, or the user can even modify the pre-installed ones, without affecting the repository layer.

Platform and libraries Then install.sh command will copy also a directory called "exe/", inside which there are a few compilation files.

In particular, "exe/astra_rc" contains the detailed list of the needed libraries and their paths.

It is good for the user to check once if these paths are correct.

If the user does not have specific libraries that are listed, these can be left blank. However, the user may need to also remove some subroutines or change the makefile "exe/Makesub" and "exe/Makexpr", which contains links to code-specific libraries.

After the "install.sh" copy is done, the user can now run ASTRA using the new command line:

exe/as.exe -m (modelfile) -v (expfile) -s (starttime) -e (endtime) -dev (machinename) -tpause (pausetime) -f -b

Note that any of these arguments is optional and does not need to be inserted. In which case, the last used value will be adopted.

The new argument "-f" launches ASTRA in "flight-simulator" mode, meaning it will crash if its not linked to the control system (whatever framework it maybe).

The argument "-dev" tells ASTRA which machine its running for configuration file purposes.

Once Astra is called, it will compile everything present in the source code and the user-defined directories sbr/ and fnc/

2 Changes in the model file syntax

A few changes have appeared in the way the model file syntax has to be given.

3 Equilibrium solvers

The variable **IPEQL**, that can be defined in the equ/log/ file in the "Parameter" block, can now attain several values, which correspond to the following equilibrium options:

"-2" * cylindrical analytical equilibrium;

"-1" * take the metric coefficients from the experimental file. The parameters (profiles) that should be given are: SHX (shift), ELX (elon), TRX (tria), G33X (G33), IPOLX (IPOL), VRX (VR), SLATX (slat), G11X (G11), G22X (G22), DRODAX (droda);

"0" * fixed guess and no equilibrium from outside;

"1" * 3-moment equilibrium EMEQ;

"2" * not used;

"3" * not used;

"4" * SPIDER prescribed boundary;

"5" * FEQIS prescribed or free boundary;

4 Generate the machine configuration grid for FEQIS

In the main Astra directory, run the following command first:

```
./Makegenerate
```

and then run the program:

`./generate_grids_equil.exe`

when prompted which machines to generate, insert 0 or 1 depending on which machine the user wants to generate.

5 Additional routines for the flight simulator

Astra can be coupled to an external control framework using a generic C interface (developed by C. Wu at KIT, Germany). It can then be run in "flight simulator" mode adding the "-f" flag in the command line.

The coupling routines are in `src/for/`, `wrsudg.f90`, `rrsudg.f90`, and `shmrw.c`. These are generic, but machine-dependent needs are specifically in `wrsudg.f90` and `rrsudg.f90`.

Additional physics files are in the `FLIGHT_SIMULATOR` directory for common or specific machines.

6 The new equilibrium solver FEQIS

The FEQIS Grad-Shafranov solver [E. Fable, Max-Planck-Institut fuer Plasmaphysik, IPP Garching, Germany, year 2023]

It is the second coupled full equilibrium solver after SPIDER [A. A. Ivanov and S. Yu. Medvedev].

It can be selected writing:

...

`IPEQL = 5.` ...

in the log file.

FEQIS, same as SPIDER, does PBE (Prescribed boundary equilibrium) and FBE (free boundary equilibrium), where FBE can also do inverse mode at the first iteration (fit of coil and passive currents).

Here the inputs of FEQIS are described, whereas the outputs are the same as SPIDER inside the ASTRA interface framework.

6.1 Input files

The input files all go into "exp/equ/(machine name)", where machine name is a 4 letter name. So far usable ones are: "aug_" for aug and "dem_" for demo. "iter" and "tcv_" are in progress.

The input files are the following:

"coil.dat": coils geometry and coil currents, same structure as SPIDER.

"coilres.dat": coils resistances same structure as SPIDER.

"data.dat": grid definition, same as SPIDER.

"limpnt.dat": limiter file, same as SPIDER.

"blanbp.dat": segmented first wall description, same as SPIDER.

"blanbpc.dat": block first wall description, new.

"efonfit.dat": fit parameters, new.

"errors_data.dat" : *error file, new.*

"refit_ef.dat" : *refit parameters, new.*

In this section the newly implemented transport equation to advance toroidal momentum is derived in details and the options to call it in the model file are presented.

The motivation that has pushed to add this important piece of physics stems from the fact that application of toroidal momentum transport theory with respect to experimental studies has rapidly expanded and it is now a very important part of the fusion research. In the previous Astra versions, it was possible to mock-up momentum transport via the F_i equations. However that has strong limitations and the physical representation of momentum transport through density equations is not accurate in most cases. That is why this equation has been added to the code.

The derivation presented here follows standard literature, from old classic derivations as in Hinton–Hazeltine review on tokamak plasma transport theory, to more recent as in V. Rozhansky *et al.* or in J. D. Callen *et al.*

6.2 Derivation of the equation implemented in Astra-7.0

The basic assumption behind the derivation of the toroidal momentum transport equation is macroscopic plasma quasi-neutrality:

$$\nabla \cdot \mathbf{J} = 0 \quad (1)$$

Its flux-surface ($\langle f \rangle \doteq \frac{\partial}{\partial V} \int f dV$) average reads

$$\langle J_\Psi R B_p \rangle = 0 \quad (2)$$

with J_r the net radial current of the bulk plasma, R the local major radius, and B_p the poloidal magnetic field.

The radial current density is given by non-ambipolar particle transport. In the case of a multi-species plasma this writes down as:

$$J_\Psi = e \sum_s Z_s \Gamma_s \quad (3)$$

with eZ_s the species charge, and Γ_s the cross-flux surface particle flux of species s .

6.3 Evaluation of the radial current and consequences on momentum transport

Each species s has its own momentum transport equation, that is the complete three-dimensional force balance:

$$\frac{\partial(m_s n_s \mathbf{U}_s)}{\partial t} + \nabla \cdot (\Pi_{c,s} + \Pi_{n,s} + \Pi_{a,s}) = eZ_s n_s (\mathbf{E} + \mathbf{U}_s \times \mathbf{B}) + \mathbf{F}_s + \mathbf{S}_s \quad (4)$$

The different quantities and terms are respectively: m_s the species mass, n_s the species density, $\Pi_{x,s}$ the species anisotropic stress tensor (c=classical, n=neoclassical, a=anomalous). Note that the anomalous part supposedly contains also the turbulent part of the Maxwell stress tensor. $\mathbf{E}, \mathbf{B}$ are the electric and magnetic fields, $\mathbf{F}_s$ the friction force, and $\mathbf{S}_s$ the externally applied torque (NBI for example).

We now multiply scalarly momtranspeq3d by $R\mathbf{e}_\phi$ and sum over all species:

$$\frac{\partial M_\phi}{\partial t} + \sum_s R\mathbf{e}_\phi \cdot \nabla \cdot (\Pi_{c,s} + \Pi_{n,s} + \Pi_{a,s}) = R\mathbf{e}_\phi \cdot (\mathbf{J} \times \mathbf{B}) + R \sum_s S_{\phi,s} \quad (5)$$

where $M_\phi = \sum_s m_s n_s R U_{\phi,s}$ is the plasma total toroidal angular momentum density. Notice that, for an axisymmetric system, $R\mathbf{e}_\phi \cdot (\nabla \cdot) \rightarrow \nabla \cdot (R\mathbf{e}_\phi \cdot)$, leading to:

$$\frac{\partial M_\phi}{\partial t} + \sum_s \nabla \cdot (R\mathbf{e}_\phi \cdot \Pi_s) = R\mathbf{e}_\phi \cdot (\mathbf{J} \times \mathbf{B}) + R \sum_s S_{\phi,s} \quad (6)$$

where $\Pi_s = \Pi_{c,s} + \Pi_{n,s} + \Pi_{a,s}$ On the other hand: $R\mathbf{e}_\phi \cdot (\mathbf{J} \times \mathbf{B}) = R J_\Psi B_p$.

Before taking the flux-surface average, we rewrite momtranspeq3drep2 in 'conservative-like' form:

$$\int_V \left[\frac{\partial M_\phi}{\partial t} + \sum_s \nabla \cdot (R \mathbf{e}_\phi \cdot \Pi_s) - R \mathbf{e}_\phi \cdot (\mathbf{J} \times \mathbf{B}) - R \sum_s S_{\phi,s} \right] dV = 0 \quad (7)$$

The volume integral spans inside a closed flux surface identified by a constant label Φ , which we choose as the toroidal flux enclosed by said surface. In that case the integral can be swapped with the time differential through the operation:

$$\int_V \frac{\partial M_\phi}{\partial t} dV = \left[\frac{\partial}{\partial t} \int_V M_\phi dV \right]_\Phi - \oint_\Phi M_\phi \mathbf{v}_\Phi \cdot \nabla V \frac{dS}{|\nabla V|} \quad (8)$$

By doing this, and differentiating then by Φ at constant time, we get the equation ($V' = \partial V / \partial \Phi$):

$$\begin{aligned} \frac{1}{V'} \frac{\partial}{\partial t} (V' \langle M_\phi \rangle) + \frac{\partial}{\partial V} \left\langle \left(R \sum_s \Pi_{\phi\Psi,s} - M_\phi \mathbf{v}_\Phi \cdot \frac{\nabla V}{|\nabla V|} \right) |\nabla V| \right\rangle \\ - \langle R J_\Psi B_p \rangle - \left\langle R \sum_s S_{\phi,s} \right\rangle = 0 \end{aligned} \quad (9)$$

Therefore, imposing condition (2), we end up with the relevant equation:

$$\begin{aligned} \frac{1}{V'} \frac{\partial}{\partial t} (V' \langle M_\phi \rangle) + \frac{\partial}{\partial V} \left\langle \left(R \sum_s \Pi_{\phi\Psi,s} - M_\phi \mathbf{v}_\Phi \cdot \frac{\nabla V}{|\nabla V|} \right) |\nabla V| \right\rangle = \\ = \left\langle R \sum_s S_{\phi,s} \right\rangle \end{aligned} \quad (10)$$

The equation can also be written in terms of the coordinate $\rho = \sqrt{\Phi / (\pi B_0)}$ that is used in ASTRA ($V' = \partial V / \partial \rho$), and $B_0 = B_{0v}$ is the reference **BTOR** field:

$$\begin{aligned} \frac{1}{V'} \frac{\partial}{\partial t} (V' \langle M_\phi \rangle) + \frac{1}{V'} \frac{\partial}{\partial \rho} \left[V' \left\langle R \sum_s \Pi_{\phi\Psi,s} |\nabla \rho| \right\rangle - V' \rho \frac{\dot{B}_0}{2B_0} \langle M_\phi \rangle \right] = \\ = \left\langle R \sum_s S_{\phi,s} \right\rangle \end{aligned} \quad (11)$$

or

$$\begin{aligned} \frac{1}{V'} \frac{\partial}{\partial t} (V' \langle M_\phi \rangle) + \frac{1}{V'} \frac{\partial}{\partial \rho} \left[V' R_\phi - V' \rho \frac{\dot{B}_0}{2B_0} \langle M_\phi \rangle \right] = \\ = \left\langle R \sum_s S_{\phi,s} \right\rangle \end{aligned} \quad (12)$$

Now, we have to express $M = \langle M_\phi \rangle$ in some practical way. Since $M_\phi = R \sum_s m_s n_s U_{\phi,s}$, we write $U_{\phi,s} = R \Omega_s$, where Ω_s is the angular rotation frequency.

We have to find a way to express the multispecies velocity, and to express the rotation as a flux function.

Let us start from (for a generic species): $\mathbf{U} = U_\parallel \mathbf{b} + \mathbf{U}_\perp$, and $\mathbf{U} = U_\phi \mathbf{e}_\phi + U_p \mathbf{e}_p$, where $\mathbf{U}_\perp = \frac{[\mathbf{E} - \nabla P / (Zen)] \times \mathbf{B}}{B^2}$. Note that $H = (E_r - \partial_r P / (Zen)) / (RB_p) = H_E + H_P$ (i.e. split in electric field and pressure contributions) is a flux function. Also, $U_p / B_p = K(\Psi)$ is a flux function.

So, we get:

$$U_\parallel = KB + \frac{I}{B}(H_E + H_P) \quad (13)$$

and

$$U_\phi = KB_\phi + R(H_E + H_P) \quad (14)$$

or

$$IH_E = U_\parallel B - IH_P - KB^2 \quad (15)$$

Note that $H = H(\Psi)$, i.e. is a flux function, since $H(\Psi) = - \left[\frac{\partial \Phi_0}{\partial \Psi} + \frac{1}{Zen} \frac{\partial P}{\partial \Psi} \right]$.

Let us define

$$u_\parallel = \langle U_\parallel B \rangle / B_0 = K \frac{\langle B^2 \rangle}{B_0} + \frac{I}{B_0}(H_E + H_P) \quad (16)$$

and

$$IH_E = B_0 u_\parallel - IH_P - K \langle B^2 \rangle \quad (17)$$

so that:

$$U_\phi = R \frac{B_0}{I} u_\parallel + KI \left(\frac{1}{R} - R \frac{\langle B^2 \rangle}{I^2} \right) \quad (18)$$

Then:

$$\Omega = \frac{B_0}{I} u_\parallel + KI \left(\frac{1}{R^2} - \frac{\langle B^2 \rangle}{I^2} \right) \quad (19)$$

This last equation allows to relate the code result for $u_{\parallel}$ to the experimental measurement of Ω at some location.

So, we have to compute $M = \langle M_{\phi} \rangle$, which, neglecting poloidal variations of the density (this will be upgraded in a future version), can be written as:

$$\langle R^2 \Omega \rangle = \frac{\langle R^2 \rangle B_0}{I} u_{\parallel} + KI \left(1 - \frac{\langle R^2 \rangle \langle B^2 \rangle}{I^2} \right) \quad (20)$$

In addition, for each impurity species $s \neq D$, the neoclassical formulas provide $\Delta u_{\parallel, Ds} = u_{\parallel}^D - u_{\parallel}^s$. So that we can write:

$$\sum_s m_s n_s u_{\parallel}^s = \rho_m u_{\parallel} - \sum_s m_s n_s \Delta u_{\parallel, Ds} \quad (21)$$

with the total mass density $\rho_m = \sum_s m_s n_s$. We also define $\delta_{\text{neo}} = \sum_s m_s n_s \Delta u_{\parallel, Ds}$, and $\sigma_{\text{neo}} = \sum_s m_s n_s K_s$.

We can then rewrite the average toroidal mass flow as:

$$\langle M_{\phi} \rangle = \Upsilon_0 u_{\parallel} + \Upsilon_1 + \Upsilon_2 \quad (22)$$

with

$$\begin{aligned} \Upsilon_0 &= \rho_m \frac{\langle R^2 \rangle B_{0v}}{I} \\ \Upsilon_1 &= -\frac{\langle R^2 \rangle B_{0v}}{I} \delta_{\text{neo}} \\ \Upsilon_2 &= I \left(1 - \frac{\langle R^2 \rangle \langle B^2 \rangle}{F} \right) \sigma_{\text{neo}} \end{aligned} \quad (23)$$

Note that in the limit of zero poloidal velocity and infinite friction, we can simplify $\Upsilon_1 = \Upsilon_2 = 0$.

The choice of $u_{\parallel}$ as the variable to be solved for is that the parallel direction is natural for plasma transport, as such theory-based transport model provide transport of parallel velocity. Infact, the term $\left\langle R \sum_s \Pi_{\phi\Psi, s} |\nabla \rho| \right\rangle$ can be represented by considering that:

$$\Pi_{\phi\Psi, s} \rightarrow \Pi_{\phi\Psi, s} + \frac{B_{\phi}}{B} \Pi_{\parallel\Psi, s} - \frac{B_p}{B} \Pi_{y\Psi, s} + m_s U_{\phi, s} \Gamma_s \quad (24)$$

where the parallel and binormal stress-tensors represent viscous terms at a fixed Φ surface. Also here Γ_s is the particle flux. Also:

$$R_{\phi} \rightarrow R_{\phi} + R_{\parallel} - R_y + m_s U_{\phi, s} \Gamma_s \quad (25)$$

The parallel stress tensor is in general given by (the species s is intended, and also at the moment the residual stress terms are not included):

$$\left\langle R \frac{B_\phi}{B} \Pi_{\parallel \Psi} |\nabla \rho| \right\rangle = -mnR_0^2 \langle |\nabla \rho|^2 \rangle \left[\chi \left(\frac{\partial H_e}{\partial \rho} + \frac{1}{I} \frac{\partial (IH_P)}{\partial \rho} \right) + \chi_\theta \frac{b_2}{I} \frac{\partial K}{\partial \rho} - V H_e - V_\theta \frac{b_2}{I} K - V_d H_P \right] \quad (26)$$

where χ, χ_θ are the turbulent viscosities of the toroidal and poloidal flow respectively, V, V_θ, V_d are the pinch of electric, poloidal, and diamagnetic flows. $b_2 = \langle B^2 \rangle$. The expression can be rewritten upon substituting using formulas (17), and summing over species:

$$\left\langle R \frac{B_\phi}{B} \Pi_{\parallel \Psi} |\nabla \rho| \right\rangle = \frac{R_0^2 B_0}{I} \langle |\nabla \rho|^2 \rangle \left[-a_1 \frac{\partial u_{\parallel}}{\partial \rho} + a_2 u_{\parallel} + a_3 \delta'_{\text{neo}} + a_4 \delta_{\text{neo}} + a_5 \sigma'_{\text{neo}} + a_6 \sigma_{\text{neo}} + a_7 \delta_{\text{d,neo}} \right] \quad (27)$$

with the following definitions:

$$\begin{aligned} a_1 &= \rho_m \chi \\ a_2 &= \rho_m \left(V + \chi \frac{\partial \log I}{\partial \rho} \right) \\ a_3 &= \chi \\ a_4 &= - \left(V + \chi \frac{\partial \log I}{\partial \rho} \right) \\ a_5 &= \frac{b_2}{B_0} (\chi - \chi_\theta) \\ a_6 &= \frac{b_2}{B_0} \left(V_\theta - V - \chi \frac{\partial \log I}{\partial \rho} + \chi \frac{\partial \log b_2}{\partial \rho} \right) \\ a_7 &= \left(V_d - V - \chi \frac{\partial \log I}{\partial \rho} \right) \\ \delta'_{\text{neo}} &= \sum_s m_s n_s \frac{\partial \Delta u_{\parallel, Ds}}{\partial \rho} \\ \sigma'_{\text{neo}} &= \sum_s m_s n_s \frac{\partial K_s}{\partial \rho} \\ \delta_{\text{d,neo}} &= \sum_s m_s n_s \frac{IH_P}{B_0} \end{aligned} \quad (28)$$

Of course all of the above assumes that the transport coefficients are the same for all ion species (this will be improved in a future upgrade).

The binormal stress tensor can be kept as a whole since in general cannot be represented in terms of diffusion and convection.

6.4 Internal variables

The parallel velocity to be solved for would be:

$$\text{UPAR} = u_{\parallel} \quad (29)$$

The main ions diffusivity, pinch, and residual stresses (assumed to be the same for all ion species at the moment) are given in the table below:

$$\begin{aligned} \text{XUPAR} &= \chi \\ \text{XUPAP} &= \chi_{\theta} \\ \text{CNPAR} &= V \\ \text{CNPAD} &= V_d \\ \text{CNPAP} &= V_{\theta} \\ \text{RUPAR} &= R_{\parallel} \\ \text{RUPYR} &= R_y \\ \text{RUPFR} &= R_{\phi} \\ \text{TTRQ} &= \left\langle R \sum_s S_{\phi,s} \right\rangle \end{aligned} \quad (30)$$

Table 2.1 Variables (arrays) related to the momentum transport equation

Name	Units	Definition	Description
MRHO	$\text{kg}/\text{m}^3/(10^{19} m_{\text{proton}})$	ρ_m	Total mass density $\rho_m = \sum_s m_s n_s$
DLNEO	$\text{kg}/(\text{m}^2 \text{ s})/(10^{19} m_{\text{proton}})$	δ_{neo}	$\delta_{\text{neo}} = \sum_s m_s n_s \Delta u_{\parallel, Ds}$
DLNEOD	$\text{kg}/(\text{m}^3 \text{ s})/(10^{19} m_{\text{proton}})$	δ'_{neo}	$\delta'_{\text{neo}} = \sum_s m_s n_s \frac{\partial \Delta u_{\parallel, Ds}}{\partial \rho}$
SGNEO	$\text{kg}/(\text{m}^2 \text{ s T})/(10^{19} m_{\text{proton}})$	σ_{neo}	$\sigma_{\text{neo}} = \sum_s m_s n_s K_s$
SGNEOD	$\text{kg}/(\text{m}^3 \text{ s T})/(10^{19} m_{\text{proton}})$	σ'_{neo}	$\sigma'_{\text{neo}} = \sum_s m_s n_s \frac{\partial K_s}{\partial \rho}$
DDNEO	$\text{kg}/(\text{m}^2 \text{ s})/(10^{19} m_{\text{proton}})$	δ_d	$\delta_d = \sum_s m_s n_s I H_{p,s} / B_0$
UPAR	m/s	$u_{\parallel}$	Parallel velocity as defined in equation (16)
XUPAR	m^2/s	χ	Radial diffusivity of parallel momentum
XUPAP	m^2/s	χ_{θ}	Radial diffusivity of poloidal momentum
CNPAR	m/s	V	Radial pinch of parallel momentum
CNPAD	m/s	V_d	Radial pinch of diamagnetic momentum
CNPAP	m/s	V_{θ}	Radial pinch of poloidal momentum
RUPAR	$1/\text{s}^2$	$R_{\parallel}$	Parallel Reynold stress $R_{\parallel} = \frac{\sum_s \langle R \frac{B_{\phi}}{B} \Pi_{\parallel \Psi, s} \nabla \rho \rangle}{(10^{19} m_{\text{proton}})}$
RUPYR	$1/\text{s}^2$	R_y	Perp Reynold stress $R_y = \frac{\sum_s \langle R \frac{B_p}{B} \Pi_{y \Psi, s} \nabla \rho \rangle}{(10^{19} m_{\text{proton}})}$
RUPFR	$1/\text{s}^2$	R_{ϕ}	Toroidal Reynold stress $R_{\phi} = \frac{\sum_s \langle R \Pi_{\phi \Psi, s} \nabla \rho \rangle}{(10^{19} m_{\text{proton}})}$
TTRQ	$\text{N}/\text{m}^2/(10^{19} m_{\text{proton}})$	$\langle R \sum_s S_{\phi, s} \rangle$	Toroidal input torque

Notice that MRHO, DLNEO, SGNEO, XUPAR, XUPAP, CNPAR, CNPAD, CNPAP, RUPAR, RUPYR, RUPFR, TTRQ, DLNEOD, DDNEO, SGNEOD have to be provided by the user in the model file, while UPAR is the evolved quantity.

6.5 Dedicated control parameters

Table 2.2 Control parameter **IPROT**

Name	Value	Description
IPROT	0	Include poloidal rotation terms
	1	Neglect poloidal rotation terms

By default, **IPROT** = 0.